Finite State Machines

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Finite State Machines (A17690)

Short Title:	Finite State Machines		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	MFURLONG		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

Finite state machines, also known as sequential circuits, are analysed in this module using formal models (Moore and Mealy) to define and specify the system requirements and produce a structured design approach. An appreciation of the problems generated by the absence of a master clock signal in asynchronous circuits will also be developed. Design solutions will be implemented using MSI components and simulated using an appropriate HDL.

Programmes

`FISM-0001`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 3 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	2 / 3 / M

Indicative Content

- Finite State Machines
- Synchronous Finite State Machines
- Asynchronous Finite State Machines
- Good state machine design
- HDL Implementations of Finite State Machines
- Clock switching techniques.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Define the fundamental concepts of finite state design.
2. Demonstrate the relevance of finite state machine optimisation.
3. Generate and test finite state machine implementations and ensure that critical race and hazard conditions, in particular, do not materialise in the asynchronous design options explored.
4. Implement linear sequential circuit applications.
5. Use HDL as a basis for the description and simulation of finite state machines.
6. Build, test and simulate the behaviour of finite state machines, relevant to the theory lectures, in a laboratory setting.

Learning and Teaching Methods

- Learning and teaching methods and strategies:
- 1. Lectures
- 2. Practicals /Mini Project
- 3. Integrated demonstrations and examples

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	15%	
In-Class Assessment	15%	1,2,3,4,5
Final Practical	35%	4,5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Floyd, D. _Digital Fundamentals_. 11th ed. New Jersey: Pearson Education, 2015.
- Lewin, D. and D. Protheroe. _Design of Logic Systems_. 2nd ed. London: Chapman &Hall, 1992.
- Wakerly, J. _Digital Design,Principles and Practices_. 3rd ed. New Jersey: Prentice Hall, 2005.

Supplementary Material

- Ashenden , P. _The Students Guide To VHDL_. 2nd ed. San Francisco: Morgan Kauffman, 2008.
- Katz, H. and G. Borriello . _Contemporary Logic Design_. 2nd ed. New Jersey: Prentice-Hall, 2004.
- Palnitkar, S. _Verilog HDL_. 2nd ed. New Jersey: Prentice-Hall, 2003.
- Xilinx, C. _Home Page for Programmable Logic_ . .: http://www.xilinx.com/, 2015.